3. Project Design Phase

3.1 Microservices Architecture & Topology

The architecture of the Personalized Networking Assistant is decoupled into microservices, cleanly separating the user interface, routing server, async workers, and persistent relational storage:

Figure 3.1: Microservice Architectural Mesh and Data Sync Pipeline

The web layer dispatches clean, async HTTPS requests directly to the core FastAPI container. The gateway processes basic user credentials immediately and delegates resource-intensive analytical tasks, such as running relationship decay formulas, to the decoupled background execution worker pool.

3.2 Relational Entity Schema & Vector Mappings

Data uniformity and transactional safety across the system are managed via fully structured database schema declarations:

Database Blueprint: UserProfiles Table

Data Property	Data Schema Type	Constraints & Keys Matrix	Architectural Definition
user_id	UUIDv4	PRIMARY KEY, NOT NULL	Primary unique surrogate database key.
email_address	VARCHAR(255)	UNIQUE, INDEXED	Unique account entry string for verification.
password_hash	VARCHAR(512)	NOT NULL	Cryptographically locked password hash using Argon2id.

Database Blueprint: ContactProfiles Table

Data Property	Data Schema Type	Constraints & Keys Matrix	Architectural Definition
contact_id	UUIDv4	PRIMARY KEY, NOT NULL	Unique contact reference identifier.
user_ref	UUIDv4	FOREIGN KEY → UserProfiles	Binds connection ownership to an authenticated user instance.
full_name	VARCHAR(128)	NOT NULL	Structured identity string used across client dashboards.
industry_vector	VECTOR(1536)	pgvector Indexed Bounds	Dense array tracking industry alignment profiles.
health_index	NUMERIC(5,2)	DEFAULT 100.00	Calculated numeric value evaluating network engagement.